

LATERRIERE, QC, Canada

WMO: 710400

Lat: 48.3072N

Lon: 71.1290W

Elev: 534

StdP: 14.41

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-17.4	-29.7	1.1	-21.2	-24.9	1.4	-16.8	23.4	8.3	21.7	8.3	4.7	220	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	83.9	67.1	80.4	66.1	77.3	64.7	70.5	78.8	68.5	76.7	66.7	74.7	9.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.6	103.3	75.0	65.5	96.0	72.9	63.6	89.8	71.0	34.7	79.2	33.1	76.8	31.7	74.8	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.5	16.4	DB	-29.4	89.6	5.2	2.3	-33.2	91.3	-36.2	92.7	-39.2	94.0	-43.0	95.7	(4)
(5)			WB	-29.6	74.1	5.3	2.0	-33.4	75.5	-36.5	76.7	-39.5	77.8	-43.4	79.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	5.4	8.5	19.9	35.5	50.0	59.4	64.8	62.9	55.0	42.6	29.7	14.4
	DBStd	22.76	14.26	12.60	12.02	8.47	7.66	6.53	5.23	5.40	7.45	7.70	9.24	12.31
	HDD50	6017	1384	1162	934	437	96	4	0	0	30	256	611	1103
	HDD65	10180	1849	1582	1398	884	469	192	67	103	312	696	1060	1568
	CDD50	1454	0	0	1	2	96	286	460	402	180	26	1	0
	CDD65	143	0	0	0	0	4	24	62	40	12	1	0	0
	CDH74	1488	0	0	1	2	109	346	586	323	117	4	0	0
	CDH80	319	0	0	0	0	28	89	127	58	17	0	0	0
	WSAvg	7.2	7.5	7.9	8.2	8.4	7.5	7.1	6.5	6.4	6.1	6.7	7.3	7.0
	PrecAvg	37.7	2.4	2.1	2.2	2.3	3.1	3.6	4.6	4.0	4.0	3.3	3.0	3.0
(15) Precipitation	PrecMax	48.0	4.8	4.4	5.0	4.2	7.1	6.3	12.6	9.6	8.9	5.9	5.8	5.6
	PrecMin	29.7	0.7	0.5	0.5	0.5	0.6	0.7	1.5	1.4	1.6	1.1	0.9	1.1
	PrecStd	4.1	0.9	0.9	1.0	1.0	1.3	1.3	1.8	1.6	1.4	1.1	1.0	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	38.5	56.6	68.3	84.5	87.7	87.6	85.5	81.9	72.5	58.6	43.1
		MCWB	39.7	34.7	47.9	51.2	62.2	68.8	69.2	69.6	67.9	61.2	51.6	39.8
	2%	DB	36.3	34.4	44.2	59.4	76.4	82.4	83.5	81.1	77.5	66.2	49.8	36.7
		MCWB	34.7	31.7	37.0	45.4	58.7	64.4	68.0	67.6	66.7	57.3	44.6	34.7
	5%	DB	32.2	31.8	40.2	53.3	70.9	77.7	80.2	77.4	73.1	59.8	45.4	34.0
		MCWB	30.8	29.6	34.4	42.9	55.5	63.2	66.6	65.9	63.8	52.6	41.7	32.3
	10%	DB	27.0	28.4	36.6	48.1	65.9	73.7	77.0	74.3	68.9	54.8	41.6	31.9
		MCWB	25.4	26.4	32.8	40.2	53.9	61.0	65.4	64.6	61.3	49.7	39.0	30.6
	0.4%	WB	40.2	35.4	48.6	54.2	67.0	71.8	73.7	72.0	71.0	62.5	52.6	39.7
		MCDB	42.7	37.7	53.1	65.0	76.1	83.4	82.4	80.7	78.7	70.0	58.3	42.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.1	32.4	38.3	47.3	62.3	68.2	70.6	69.5	67.8	57.9	46.1	35.1
		MCDB	36.3	34.2	42.7	57.3	74.0	77.6	78.6	77.7	75.1	65.2	48.7	36.4
	5%	WB	30.8	29.4	35.4	44.0	57.6	65.7	68.6	67.5	65.0	54.0	42.1	32.7
		MCDB	32.3	31.7	39.2	51.4	67.7	75.0	76.7	75.6	72.2	58.8	44.5	33.9
	10%	WB	25.2	26.4	32.9	41.3	54.4	63.0	66.6	65.9	62.3	50.2	39.3	30.4
		MCDB	26.9	28.5	36.2	47.2	64.4	71.2	74.6	72.7	68.3	54.1	41.8	32.0
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	19.3	19.9	18.0	22.1	21.1	19.6	19.0	19.8	15.0	12.9	14.8
		MCDBR	20.9	17.6	24.2	28.0	32.3	29.3	25.7	25.2	26.5	24.7	18.3	14.4
		MCWBR	19.3	15.7	18.1	17.6	16.8	15.3	13.0	13.7	16.2	16.4	14.3	12.9
	5% WB	MCDBR	20.4	16.3	19.5	24.2	26.8	24.4	21.4	21.2	23.4	22.9	16.3	14.3
		MCWBR	19.0	14.8	15.3	16.2	16.1	14.5	12.4	12.3	15.4	16.4	13.4	13.5
(40) Clear-Sky Solar Irradiance	taub		0.266	0.284	0.315	0.376	0.358	0.383	0.402	0.381	0.370	0.329	0.316	0.266
	taud		2.266	2.224	2.201	2.222	2.408	2.367	2.326	2.387	2.407	2.505	2.374	2.322
	Ebn at Noon		255	276	285	276	286	279	271	272	264	260	233	240
	Edh at Noon		23	31	37	41	36	37	38	35	31	24	22	20
(44) All-Sky Solar Radiation	RadAvg		416	721	1114	1413	1632	1742	1649	1404	1078	619	338	276
	RadStd		51	94	92	111	158	117	163	93	81	62	35	29

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page